

Data Warehouse using PostgreSQL

Assignment

The assignment is to be carried out in pairs

1. Problem Statement

A retail company sells products through physical stores, an online platform, and third-party partners. Sales data is stored in different systems and formats. Management wants all sales data consolidated into a central PostgreSQL database to analyze revenue, product performance, and customer behavior.

The warehouse should enable analysis of:

- Revenue trends
- Product performance
- Customer purchasing behaviour
- Sales across different channels
- Regional performance
- Monthly and yearly sales

1. The in-store sales is in CSV format. Contains point-of-sale transactions. The sample is attached (sales.csv).

The Dates are stored as text and there is a possibility of duplicate records

2. The online platform has a file in json format. The following is an example of a record in such a dataset.

```
{
  "order_id": "O123",
  "order_timestamp": "2025-01-15T10:45:00",
  "product_code": "P101",
  "units_sold": 3,
  "price_per_unit": 1200,
  "customer_email": "Mueni@comp.com"
}
```

Though the online platform has similar information as the physical store the column names are different for instance Timestamp instead of date.

3. The partner sales data is in Excel and includes fields such as:

transaction_no, date_of_sale, item_id, total_amount and partner_name

Notably, the quantity is not explicitly given and has the column total_amount instead of unit price.

Students are expected to identify suitable datasets that resemble online sales and partner sales while ensuring that the attributes can be integrated into a common warehouse schema.

2. Software Requirements

- PostgreSQL 15 or later
- pgAdmin
- Python (optional for ETL)
- Microsoft Excel
- Any JSON editor

3. Requirements

You are tasked with building an ETL pipeline to achieve this, namely:

1. Describe and explore each of the datasets. Identify any preprocessing required before integration.
 2. Design a dimensional model. Draw the resulting ER/Data Warehouse diagram
 3. Implement a star or snowflake schema in PostgreSQL Write the SQL codes for each of the table.
 4. Integrate heterogeneous datasets.
 5. Load data into fact and dimension tables.
 6. Optimize warehouse performance using partitioning, indexing and materialized views
 7. Perform multidimensional analysis using SQL: OLAP operations such as drill-down, roll-up, slice and dice.
 8. Performance Evaluation: Query performance before and after optimization in terms of execution time, query plan and improvements observed.
4. Deliverables

Each pair should submit:

1. The two additional datasets (JSON and Excel).
2. A dimensional model (Star or Snowflake Schema).
3. SQL scripts for database creation.
4. SQL scripts for ETL/loading.
5. SQL scripts implementing partitioning, indexing, and materialized views.
6. SQL scripts for all OLAP queries.
7. Performance comparison using EXPLAIN ANALYZE.
8. A report (5–8 pages) discussing:
 - Schema design decisions
 - ETL process
 - Optimization techniques
 - OLAP analysis
 - Performance evaluation
 - Conclusions and recommendations.